QUANTITATIVE TORIC FLUX-DISTANCE RIGIDITY AND GEOMETRIC BOUNDEDNESS OF CONVEX TORUS DOMAINS

YOEL GROMAN

Allowed standard prerequisites (editorial).

- Symplectic cutting, holomorphic-curve compactness and standard toric/Fukaya background as used by author. Original Sikorav, STVFluxShape and FOOO/ChoOh bibliography retained; actual source-local arguments included.

Source reference. Original reference 1.22: algebra-geometry/sources/sigma-2026-066/native/sigma26-066.tex, source lines 381–381. Later application or generalization mentioned in the narrative; not an input to the retained target proof.

0.1. **The results.** As our first proxy for completeness, we use the well-established notion of *geometric boundedness* (see, e.g., [14, Section 4.2]): an ω-compatible almost complex structure J on M is called *geometrically bounded* if the associated Riemannian metric $g_J(\cdot,\cdot) = \omega(\cdot, J\cdot)$ is complete and has uniformly bounded sectional curvature and uniformly positive injectivity radius.[1]

We begin with an existence question.

Question 0.1. Given ω, does there exist a geometrically bounded ω-compatible almost complex structure J on M?

It is customary to call (M,ω) itself geometrically bounded if the answer to Question 0.1 is affirmative. We consider this question when we start with a geometrically bounded M and excise a subset $N \subset M$. Question 0.1 can be understood as the question: *can the distance to N of any compact subset of $M \setminus N$ be made arbitrarily large by adjusting the almost complex structure?*

Theorem 0.2. *Let M be a geometrically bounded symplectic manifold with finitely generated spherical homology, and suppose ω is rational. Let $N \subset M$ be a properly embedded symplectic hypersurface. Then $M \setminus N$ is not geometrically bounded.*

Theorem 0.3. *Let $U \subset \mathbb{R}^n$ be convex. Let ω be the standard symplectic form on $\mathbb{R}^n \times \mathbb{T}^n$. If there exists an ω-compatible almost complex structure on $U \times \mathbb{T}^n$ which is geometrically bounded, then $U = \mathbb{R}^n$.*

Theorem 0.3 follows from the following more quantitative statement which is of independent interest. Identify primitive classes in $H_1(\mathbb{T}^n;\mathbb{Z})$ with primitive vectors $v \in \mathbb{Z}^n$, hence with integral conormals. For an open convex set $U \subset \mathbb{R}^n$ and $p \in U$, let $L_p := \{p\} \times \mathbb{T}^n \subset U \times \mathbb{T}^n$. Define the "flux distance" to the boundary by

$$\delta_{\mathrm{flux}}(p;\partial U) := \inf_{v \in \mathbb{Z}^n \,\mathrm{primitive}} \sup_{x \in \partial U} \langle v, p - x\rangle \in (0, +\infty).$$

Proposition 0.4. *Fix $a > 0$. There exists a universal dimensionless constant $C > 0$ such that for every open convex $U \subset \mathbb{R}^n$, every $p \in U$, and every ω-compatible almost complex structure J on $\mathbb{R}^n \times \mathbb{T}^n$ for the standard symplectic form*

Date: doi:10.3842/SIGMA.2026.066; CC BY 4.0.

[1] To treat ω-tame almost complex structures, one must also impose a uniform bound on the norm of ω with respect to g_J (i.e., $|\omega|_{g_J} \le C$).

with g_J a-bounded (see Definition 1.3 below), one has

$$d_{g_J}(L_p, (\mathbb{R}^n \setminus U) \times \mathbb{T}^n) \leq Ca\delta_{\text{flux}}(p; \partial U)$$

provided $\delta_{\text{flux}}(p; \partial U) \gg 1/a$.

This is proved in Section 1 and immediately implies Theorem 0.3.

1. Proofs

1.1. **Energy confinement for J-holomorphic curves.** Throughout, if $u\colon (\Sigma, \partial\Sigma) \to (M, \omega)$ is a J-holomorphic curve with (possibly empty) boundary, its energy is $E(u) := \int_\Sigma u^*\omega \geq 0$.

The following confinement lemmas are stated for ω-compatible J. At the end of this subsection we note that both hold for ω-tame J under the uniform tameness condition from the footnote to the definition of geometric boundedness on page 1 (in Section 1.1).

Lemma 1.1 (Sikorav's confinement estimate)**.** *Let (M, ω) be a symplectic manifold and let J be an ω-compatible almost complex structure such that the associated metric $g_J(\cdot, \cdot) := \omega(\cdot, J\cdot)$ is complete and has uniformly bounded sectional curvature and positive injectivity radius (i.e., J is geometrically bounded). Then there exist a constant $r_0 > 0$ depending on the geometry and a universal constant $c > 0$ such that for every connected J-holomorphic curve u one has*

$$u(\Sigma) \subset B_{r_0 \lceil E(u)/(cr_0^2) \rceil}(u(\partial\Sigma)),$$

where $B_r(\cdot)$ denotes the closed g_J-ball of radius r about a subset and $\lceil\cdot\rceil$ is the ceiling function.

Proof. See [14, Proposition 4.4.1]. □

This confinement estimate is concerned with the large scale. Namely, it contains no information about curves with energy much smaller than cr_0^2 other than that the diameter is $\leq r_0$. In some cases, e.g., in the proof of Proposition 1.22, we need an estimate for the small scale. This is taken care of by the following lemma.

Lemma 1.2 (improved confinement)**.** *In the setting of Lemma 1.1, with the same r_0 and c, and letting $C := 1/c$, one has*

$$u(\Sigma) \subset B_\rho(u(\partial\Sigma)), \qquad \rho := \max\bigl(3CE(u)/r_0, \sqrt{CE(u)}\bigr).$$

Proof. The argument relies on the same monotonicity inequality used for Sikorav's lemma: if a J-holomorphic curve meets the center of a ball of radius $r \leq r_0$ and the boundary of the J-holomorphic curve does not meet the interior of the ball, then the energy in that ball is at least cr^2.

If $E(u)/\bigl(cr_0^2\bigr) \leq 1$, we apply the monotonicity inequality directly: if the image contained a point at the center of a ball of radius greater than $\sqrt{E(u)/c}$ whose interior did not meet $u(\partial\Sigma)$, then monotonicity would force $E(u) \geq cr^2 > E(u)$ (since $E(u) \geq cr^2$ implies $r \leq \sqrt{E(u)/c}$). Hence $u(\Sigma)$ is contained in a ball of radius $\sqrt{E(u)/c} = \sqrt{CE(u)}$ about $u(\partial\Sigma)$.

If $E(u)/\bigl(cr_0^2\bigr) > 1$, take the maximum number n of disjoint balls of radius r_0 that the image of u can meet. Then $n \leq E(u)/\bigl(cr_0^2\bigr)$ by monotonicity, and the image is contained in a ball of radius $(2n+1)r_0 \leq r_0\bigl(1 + 2E(u)/\bigl(cr_0^2\bigr)\bigr) < 3E(u)/(cr_0) = 3CE(u)/r_0$ around $u(\partial\Sigma)$. □

For later use, we also record a localized version, originally observed in [6].

Definition 1.3. For $a > 0$, we say that the complete Riemannian metric g_J is *a-bounded at a point $p \in M$* if $\operatorname{inj}_{g_J}(p) \geq \frac{1}{a}$ and $\sec_{g_J} \leq a^2$ on $B_{1/a}(p)$. We say that g_J is *a-bounded on a set* if it is a-bounded at every point of that set.

Remark 1.4. When g_J is a-bounded, we can take $r_0 = 1/(2a)$ [12, Theorem 4.4.1 and Remark 4.4.3].

Lemma 1.5 (local confinement)**.** *Fix $a > 0$. There exist $r_0 = r_0(a) > 0$ and a universal constant $C > 0$ with the following property. Let $K \subset M$ be compact and let $R > 0$. Assume J is ω-compatible and g_J is complete and is a-bounded on the g_J-ball $B_R(K)$. Let u be a J-holomorphic curve u with $u(\partial\Sigma) \subset K$ and write*

$$\rho := \max\bigl(2CE(u)/r_0, \sqrt{CE(u)}\bigr).$$

If $\rho \le R$, then $u(\Sigma) \subset B_R(K)$.

Proof. The proof of Lemma 1.2 relies on the monotonicity inequality which depends only on the local geometry of a complete Riemannian metric. So taking r_0 and C as in Lemma 1.2, we obtain by the same argument that $u(\Sigma) \subset B_\rho(u(\partial\Sigma))$. □

1.2. **Proof of Theorem 0.3.** Let X be a closed symplectic toric manifold with moment polytope $P \subset \mathbb{R}^n$ and moment map $\mu\colon X \to P$. For $p \in \operatorname{Int} P$, let $L_p := \mu^{-1}(p) \cong \mathbb{T}^n$. A *basic* (*Maslov index* 2) *class* in $\pi_2(X, L_p)$ is a relative class β_F corresponding to a facet F of P. Geometrically, β_F can be constructed as follows. Let $v_F \in \mathbb{Z}^n$ be the primitive integral inward-pointing normal vector to F. The circle action generated by v_F acts freely on the interior of P but collapses to a point over the facet F. Thus, if we draw a line segment L in the moment polytope P from p to a point in the interior of F and, transporting along the orbit of this action as one traverses L, we obtain a disc in the class β_F.

Proposition 1.6. *For any $p \in \operatorname{Int} P$ and any ω-tame almost complex structure J on X, there exists a stable J-holomorphic map with boundary on L_p whose domain has at least one disc component $u_D\colon (D, \partial D) \to (X, L_p)$ of positive symplectic area such that*

$$\int_D u^*\omega \le \min_F \omega(\beta_F),$$

where the minimum is taken over facets F of P.

Remark 1.7. The inequality above is stated for the *disc component* u_D: The configuration guaranteed by the proposition may contain sphere bubbles. In that case, the disc component has strictly smaller area than $\omega(\beta_F)$.

Proof. We use the Ψ-invariant of a Lagrangian submanifold considered by Shelukhin–Tonkonog–Vianna [13]. Fix an ω-tame almost complex structure J and auxiliary choices (virtual perturbations) defining the Fukaya A_∞-algebra of L_p, and pass to a classically minimal model. The quantity $\Psi(L_p) \in (0, +\infty]$ is defined as an infimum of valuations of the Maurer–Cartan prepotential $m\bigl(e^b\bigr)$; it is independent of all choices (in particular of J) by [13, Theorem 4.2]. Furthermore, expanding the definition, $\Psi(L_p)$ is the minimum of the symplectic areas $\omega(\beta)$ over those relative classes $\beta \in H_2(X, L_p; \mathbb{Z})$ which contribute nontrivially to some symmetrized A_∞ operation; see [13, Theorem 4.8]. Note this includes the classes which contribute to the curvature term.

For any such class β contributing to the curvature term there exists a stable J-holomorphic map u representing β with at least one nonconstant disc component.[2]

[2]To see this, note that otherwise β would be a purely spherical class. The invariance of the indicator function [13, Proposition 5.1] would then dictate that it must also realize the ψ invariant for J_{toric}. However, for J_{toric}, all non-constant holomorphic spheres are confined to the toric boundary divisor and are strictly disjoint from the interior fiber L_p, so their contributions to the A_∞-algebra of L_p necessarily vanish. Hence, $\partial\beta \neq 0$.

The class area decomposes as a sum of the symplectic areas of the disc and sphere components of u, so every nonconstant disc component u_D satisfies

$$\int_D u_D^*\omega \leq \omega(\beta),$$

with strict inequality whenever u also has a nonconstant sphere component.

Now take the standard toric almost complex structure J_{toric}. Fukaya–Oh–Ohta–Ono [4] (and earlier Cho–Oh [2]) compute the leading-order terms of the toric superpotential/potential function: for each facet F the basic Maslov index 2 class β_F contributes nontrivially to the curvature term (equivalently, to the potential), with coefficient 1; see [4, Section 11] and [2]. Therefore, for a facet F minimizing $\omega(\beta_F)$, the class β_F witnesses $\Psi(L_p)$ for J_{toric}, and hence

$$\Psi(L_p) \leq \omega(\beta_F) = \min_{F'} \omega(\beta_{F'}).$$

Since $\Psi(L_p)$ is independent of J, the same inequality holds for our given J. Applying [13, Theorem 4.8] for this J, we obtain a stable J-holomorphic map representing some class β with $\omega(\beta) = \Psi(L_p) \leq \min_{F'} \omega(\beta_{F'})$. Choosing a disc component u_D of this stable map, the discussion above gives

$$\int_D u_D^*\omega \leq \omega(\beta) = \Psi(L_p) \leq \min_{F'} \omega(\beta_{F'}),$$

which is the claimed inequality. □

To prove Theorem 0.3, it suffices to prove Proposition 0.4 from the introduction.

Write $\pi\colon \mathbb{R}^n \times \mathbb{T}^n \to \mathbb{R}^n$ for the projection, and for $q \in \mathbb{R}^n$ write $L_q := \{q\} \times \mathbb{T}^n \subset \mathbb{R}^n \times \mathbb{T}^n$. Recall the definition of $\delta_{\text{flux}}(p; \partial U)$ from the introduction. For later use, given $v \in \mathbb{Z}^n$ we write

$$c_{\partial U}(v) := \inf_{x \in \partial U} \langle v, x\rangle \in [-\infty, +\infty), \qquad \delta_{\partial U}(p; v) := \langle v, p\rangle - c_{\partial U}(v) \in (0, +\infty],$$

so $\delta_{\partial U}(p; v) < +\infty$ precisely when $c_{\partial U}(v) > -\infty$.

We will deduce Theorem 0.3 and Proposition 0.4 from the following stronger version of Proposition 0.4.

Proposition 1.8. *Let $U \subset \mathbb{R}^n$ be an open convex set, and let V be an open neighborhood of $U \times \mathbb{T}^n$ in $\mathbb{R}^n \times \mathbb{T}^n$. Let J be an ω-tame almost complex structure on V such that the associated metric g_J is complete on V and is a-bounded on $U \times \mathbb{T}^n$ for some $a > 0$. Then for every $p \in U$,*

$$d_{g_J}(L_p, V \setminus (U \times \mathbb{T}^n)) \leq \max\bigl(4Ca\delta_{\text{flux}}(p; \partial U), \sqrt{C\delta_{\text{flux}}(p; \partial U)}\bigr),$$

where C is a universal constant.

To prove Proposition 1.8, we formulate a basic lemma.

Lemma 1.9. *Let $U \subset \mathbb{R}^n$ be an open convex set, let $K \subset U$ be compact, and let $v \in \mathbb{Z}^n$ be primitive with $c_{\partial U}(v) > -\infty$. Then there exists a compact Delzant polytope P such that $K \subset \operatorname{Int} P \subset U$ and such that one facet of P has outward primitive normal v (equivalently, is parallel to the supporting hyperplane $\{\langle v, \cdot\rangle = c_{\partial U}(v)\}$).*

Proof. Choose $\varepsilon > 0$ such that the closed ε-neighborhood $K_\varepsilon := \{x \mid d(x, K) \leq \varepsilon\}$ is still contained in U (possible since U is open and K is compact). Pick $c < c_{\partial U}(v)$ close enough to $c_{\partial U}(v)$ so that $K_\varepsilon \subset \{x \mid \langle v, x\rangle > c\}$, and set $H := \{x \mid \langle v, x\rangle = c\}$ and $H^- := \{x \mid \langle v, x\rangle \geq c\}$.

Since K_ε is compact, we can find finitely many supporting half-spaces with *rational* (hence integral after scaling) outward normals whose intersection contains K_ε and is contained in U. Intersecting these half-spaces together with H^- yields

a compact simple rational polytope P_0 with $K \subset \operatorname{Int} P_0 \subset U$ and with a facet parallel to H.

Finally, by applying arbitrarily small *corner cuts* one can make the polytope smooth/Delzant while keeping the containments $K \subset \operatorname{Int} P \subset U$ and without changing the facet with outward normal v. □

Proof of Proposition 1.8. Let $\delta := \delta_{\mathrm{flux}}(p; \partial U)$. Fix $\varepsilon > 0$, and choose a primitive $v \in \mathbb{Z}^n$ with $c_{\partial U}(v) > -\infty$ and $\delta_{\partial U}(p; v) \le \delta + \varepsilon$. Put $c := c_{\partial U}(v)$ and $\ell(x) := \langle v, x\rangle - c$, so $\ell(p) = \delta_{\partial U}(p; v)$.

Let J be as in the statement. We will prove the inequality

$$d_{g_J}(L_p, V \setminus (U \times \mathbb{T}^n)) \le \max\bigl(4Ca\delta_{\mathrm{flux}}(p; \partial U), \sqrt{C\delta_{\mathrm{flux}}(p; \partial U)}\bigr). \tag{1.1}$$

Let $R > 0$ be such that the closed metric ball $B := B_R(L_p)$ with respect to g_J is contained in $U \times \mathbb{T}^n$. Since g_J is complete on V and $B \subset U \times \mathbb{T}^n \subset V$, the closed ball B is compact. In particular, $\pi(B) \subset U$ is compact.

Apply Lemma 1.9 to $K := \pi(B) \subset U$ and the chosen v to obtain a Delzant polytope P with $\pi(B) \subset \operatorname{Int} P \subset U$ and with a facet $F \subset \partial P$ whose outward primitive normal is v. Let (X_P, ω_P, μ_P) be the associated compact symplectic toric manifold with moment polytope P. By the action–angle identification, $\mu_P^{-1}(\operatorname{Int} P)$ is symplectomorphic to $\operatorname{Int} P \times \mathbb{T}^n$; we use this to view B as a subset of X_P. Extend $J|_B$ arbitrarily to an ω_P-tame almost complex structure $\overline{J}$ on X_P.

Let $A := \min_{F'} \omega_P(\beta_{F'})$, where F' ranges over facets of P. Since F is one of the facets, we have $A \le \omega_P(\beta_F)$. Moreover, because $P \subset U$ and the facet F has outward normal v, the defining constant c_P of that facet satisfies $c_P \ge c_{\partial U}(v) = c$, hence

$$\omega_P(\beta_F) = \langle v, p\rangle - c_P \le \langle v, p\rangle - c = \delta_{\partial U}(p; v) \le \delta + \varepsilon.$$

By Proposition 1.6 applied to (X_P, ω_P, L_p) and $\overline{J}$, there exists a stable $\overline{J}$-holomorphic map with a disc component u_D and $\partial u_D \subset L_p$ satisfying

$$E(u_D) = \int_D u_D^* \omega_P \le A \le \delta + \varepsilon.$$

The disc component u_D must meet the toric boundary $X_P \setminus \mu_P^{-1}(\operatorname{Int} P) = \mu_P^{-1}(\partial P)$. Indeed, one can pick a primitive of ω_P on $\mu_P^{-1}(\operatorname{Int} P)$ for which L_p is exact, showing that any disc with boundary on L_p with positive symplectic area must meet the toric boundary.

On the other hand, since $g_{\overline{J}}$ is a-bounded on $B = B_R(L_p)$, Lemma 1.5 implies that $u_D(D)$ is contained in the ball of radius $\rho = \max\bigl(2CE(u_D)/r_0, \sqrt{CE(u_D)}\bigr)$ about L_p, with $r_0 = r_0(a)$ and the universal constant C from that lemma. Since u_D meets $\mu_P^{-1}(\partial P)$ while, by construction, $B_R(L_p)$ does not, we must have $R \le \rho \le C(a) \max\bigl(\delta + \varepsilon, \sqrt{\delta + \varepsilon}\bigr)$. Since $\varepsilon > 0$ was arbitrary, this gives the claim. □

Proof of Proposition 0.4. Apply Proposition 1.8 with $V = \mathbb{R}^n \times \mathbb{T}^n$. Since J is ω-compatible and g_J is a-bounded globally on $\mathbb{R}^n \times \mathbb{T}^n$, it is a-bounded on $U \times \mathbb{T}^n$. Moreover, g_J is complete on V. The conclusion of Proposition 1.8 exactly yields the desired inequality. □

Proof of Theorem 0.3. Suppose for contradiction that $U \neq \mathbb{R}^n$. Since U is open and convex, its boundary is nonempty, so $\delta_{\mathrm{flux}}(p; \partial U)$ is finite for any $p \in U$. The hypothesis states that there exists a geometrically bounded ω-compatible almost complex structure J on $U \times \mathbb{T}^n$. This means g_J is complete on $U \times \mathbb{T}^n$ and is a-bounded for some $a > 0$. Apply Proposition 1.8 with $V = U \times \mathbb{T}^n$. Since $V \setminus (U \times \mathbb{T}^n) = \varnothing$, the distance $d_{g_J}(L_p, V \setminus (U \times \mathbb{T}^n))$ is infinite. However, Proposition 1.8 provides a finite upper bound in terms of $\delta_{\mathrm{flux}}(p; \partial U)$, which is a contradiction. Hence $U = \mathbb{R}^n$. □

I used AI-assisted tools for editing and improving the exposition (in particular for wording, structure, and catching typographical issues). All mathematical content, results, and any remaining errors are my own responsibility.

References

[1] Auroux D., Mirror symmetry and T-duality in the complement of an anticanonical divisor, *J. Gökova Geom. Topol. GGT* **1** (2007), 51–91, arXiv:0706.3207.

[2] Cho C.-H., Oh Y.-G., Floer cohomology and disc instantons of Lagrangian torus fibers in Fano toric manifolds, *Asian J. Math.* **10** (2006), 773–814, arXiv:math.SG/0308225.

[3] Entov M., Ganor Y., Membrez C., Lagrangian isotopies and symplectic function theory, *Comment. Math. Helv.* **93** (2018), 829–882, arXiv:1610.09631.

[4] Fukaya K., Oh Y.-G., Ohta H., Ono K., Lagrangian Floer theory on compact toric manifolds. I, *Duke Math. J.* **151** (2010), 23–174, arXiv:0802.1703.

[5] Gallot S., Hulin D., Lafontaine J., Riemannian geometry, 3rd ed., *Universitext*, Springer, Berlin, 2004.

[6] Groman Y., Floer theory and reduced cohomology on open manifolds, *Geom. Topol.* **27** (2023), 1273–1390, arXiv:1510.04265.

[7] Groman Y., The wrapped Fukaya category for semi-toric Calabi–Yau, *J. Eur. Math. Soc. (JEMS)* **26** (2024), 2373–2439, arXiv:1805.03635.

[8] Groman Y., Varolgunes U., Locality of relative symplectic cohomology for complete embeddings, *Compos. Math.* **159** (2023), 2551–2637, arXiv:2110.08891.

[9] Günther P., Einige Sätze über das Volumen von geodätischen Kugeln, *Ber. Verh. Sächs. Akad. Wiss. Leipzig Math.-Nat. Kl.* **104** (1960), 1–38.

[10] Karshon Y., Tang X., Symplectic excision, arXiv:2101.03534.

[11] McDuff D., Symplectic structures – a new approach to geometry, *Notices Amer. Math. Soc.* **45** (1998), 952–960.

[12] McDuff D., Salamon D., J-holomorphic curves and symplectic topology, 2nd ed., *Amer. Math. Soc. Colloq. Publ.*, Vol. 52, American Mathematical Society, Providence, RI, 2012.

[13] Shelukhin E., Tonkonog D., Vianna R., Geometry of symplectic flux and Lagrangian torus fibrations, *J. Topol.* **17** (2024), e70002, 56 pages, arXiv:1804.02044.

[14] Sikorav J.-C., Some properties of holomorphic curves in almost complex manifolds, in Holomorphic Curves in Symplectic Geometry, *Progr. Math.*, Vol. 117, Birkhäuser, Basel, 1994, 165–189.

[15] Stratmann B., Removing parametrized rays symplectically, *J. Symplectic Geom.* **20** (2022), 499–508, arXiv:2009.05465.